1 Wavenumber Integration Formulation

We consider acoustic propagation in a horizontally stratified (range-independent) environment, where the sound speed $c(z)$ and density $\rho(z)$ vary with depth z only. The depth axis points downwards, with the sea surface at $z = 0$. An acoustic source located at $\boldsymbol{r}_s = (0, 0, z_s)$ omni-directionally transmits a CW signal at frequency f.

The acoustic pressure is governed by the wave equation

$$\frac{\partial^2 p}{\partial t^2} = c^2 \nabla^2 p. \tag{1}$$

Which has a variable-separated solution

$$p(\boldsymbol{r}, t) = \bar{p}(\boldsymbol{r})\, e^{-i\omega t}, \tag{2}$$

$$\frac{\partial^2}{\partial t^2}\left[\bar{p}(\boldsymbol{r})\, e^{-i\omega t}\right] = -\omega^2\, \bar{p}(\boldsymbol{r})\, e^{-i\omega t}$$

$$c^2 \nabla^2 \left[\bar{p}(\boldsymbol{r})\, e^{-i\omega t}\right] = c^2\, e^{-i\omega t}\, \nabla^2 \bar{p}(\boldsymbol{r})$$

$$-\omega^2\, \bar{p}(\boldsymbol{r})\, e^{-i\omega t} = c^2\, e^{-i\omega t}\, \nabla^2 \bar{p}(\boldsymbol{r})$$

$$-\frac{\omega^2}{c^2(z)}\, \bar{p}(\boldsymbol{r}) = \nabla^2 \bar{p}(\boldsymbol{r})$$

$$k^2(z)\, \bar{p}(\boldsymbol{r}) + \nabla^2 \bar{p}(\boldsymbol{r}) = 0$$

$$k^2(z)\, \bar{p}(\boldsymbol{r}) + \nabla^2 \bar{p}(\boldsymbol{r}) = -\delta(\boldsymbol{r} - \boldsymbol{r}_s), \qquad k(z) = \frac{\omega}{c(z)}. \tag{3}$$

Which is the Helmholtz's equation whose solution in an unbounded homogeneous medium is Green's function of an outgoing spherical wave

$$\bar{p}(\boldsymbol{r}) = \frac{e^{\,ik\|\boldsymbol{r} - \boldsymbol{r}_s\|_2}}{4\pi\, \|\boldsymbol{r} - \boldsymbol{r}_s\|_2} \tag{4}$$

However, this free-field solution is unbounded and cannot be fitted to the boundary conditions of the stratified waveguide.

Since both the environment and the source are rotationally symmetric about the vertical axis through the source, we switch to cylindrical coordinates:

$$\nabla^2 \bar{p} = \frac{1}{r}\frac{\partial}{\partial r}\left(r \frac{\partial \bar{p}}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 \bar{p}}{\partial \phi^2} + \frac{\partial^2 \bar{p}}{\partial z^2}$$

$$\frac{\partial \bar{p}}{\partial \phi} = 0 \quad \Longrightarrow \quad \nabla^2 \bar{p} = \frac{1}{r}\frac{\partial}{\partial r}\left(r \frac{\partial \bar{p}}{\partial r}\right) + \frac{\partial^2 \bar{p}}{\partial z^2}$$

$$\delta(\boldsymbol{r} - \boldsymbol{r}_s) = \frac{\delta(r)\, \delta(z - z_s)}{2\pi r}$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\,\frac{\partial \bar{p}}{\partial r}\right)+\frac{\partial^2 \bar{p}}{\partial z^2}+k^2(z)\,\bar{p}(r,z)=-\,\frac{\delta(r)\,\delta(z-z_s)}{2\pi r}. \tag{5}$$

Now we continue to define the two-dimensional transform pair

$$\tilde{p}(k_x,k_y,z)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\bar{p}(x,y,z)\,e^{-i(k_x x+k_y y)}\,dx\,dy, \tag{6}$$

$$\bar{p}(x,y,z)=\frac{1}{(2\pi)^2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\tilde{p}(k_x,k_y,z)\,e^{i(k_x x+k_y y)}\,dk_x\,dk_y. \tag{7}$$

Using the identity $\int_0^{2\pi} e^{-ik_r r\cos\varphi}\,d\varphi = 2\pi J_0(k_r r)$, the pair (6)–(7) reduces to the zeroth-order Hankel transform pair

$$x=r\cos\theta,\quad y=r\sin\theta,\qquad k_x=k_r\cos\alpha,\quad k_y=k_r\sin\alpha\quad\Longrightarrow\quad k_x x+k_y y=k_r r\cos(\theta-\alpha)$$

$$\tilde{p}(k_r,z)=\int_0^{\infty}\int_0^{2\pi}\bar{p}(r,z)\,e^{-ik_r r\cos(\theta-\alpha)}\,r\,d\theta\,dr=2\pi\int_0^{\infty}\bar{p}(r,z)\,J_0(k_r r)\,r\,dr$$

$$g(k_r,z)\equiv\frac{\tilde{p}(k_r,z)}{2\pi}=\int_0^{\infty}\bar{p}(r,z)\,J_0(k_r r)\,r\,dr$$

$$\bar{p}(r,z)=\frac{1}{(2\pi)^2}\int_0^{\infty}\int_0^{2\pi}2\pi\,g(k_r,z)\,e^{ik_r r\cos(\alpha-\theta)}\,k_r\,d\alpha\,dk_r=\int_0^{\infty}g(k_r,z)\,J_0(k_r r)\,k_r\,dk_r$$

$$g(k_r,z)=\int_0^{\infty}\bar{p}(r,z)\,J_0(k_r r)\,r\,dr, \tag{8}$$

$$\bar{p}(r,z)=\int_0^{\infty}g(k_r,z)\,J_0(k_r r)\,k_r\,dk_r, \tag{9}$$

where $g(k_r,z)\equiv\tilde{p}(k_x,k_y,z)/(2\pi)$ and $J_0(\cdot)$ denotes the Bessel function of the first kind of order zero.

Applying (8) to (5)

$$\int_0^{\infty}\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\,\frac{\partial \bar{p}}{\partial r}\right)+\frac{\partial^2 \bar{p}}{\partial z^2}+k^2(z)\,\bar{p}(r,z)\right]J_0(k_r r)\,r\,dr=\int_0^{\infty}\left[-\,\frac{\delta(r)\,\delta(z-z_s)}{2\pi r}\right]J_0(k_r r)\,r\,dr$$

$$\int_0^{\infty}\frac{1}{r}\frac{\partial}{\partial r}\left(r\,\frac{\partial \bar{p}}{\partial r}\right)J_0(k_r r)\,r\,dr=\int_0^{\infty}\frac{\partial}{\partial r}\left(r\,\frac{\partial \bar{p}}{\partial r}\right)J_0(k_r r)\,dr$$

$$=\left[r\,\frac{\partial \bar{p}}{\partial r}\,J_0(k_r r)\right]_0^{\infty}-\int_0^{\infty}r\,\frac{\partial \bar{p}}{\partial r}\,\frac{dJ_0(k_r r)}{dr}\,dr=-\int_0^{\infty}r\,\frac{\partial \bar{p}}{\partial r}\,\frac{dJ_0(k_r r)}{dr}\,dr$$

$$=-\left[r\,\frac{dJ_0(k_r r)}{dr}\,\bar{p}\right]_0^{\infty}+\int_0^{\infty}\bar{p}\,\frac{d}{dr}\left(r\,\frac{dJ_0(k_r r)}{dr}\right)dr=\int_0^{\infty}\bar{p}\,\frac{d}{dr}\left(r\,\frac{dJ_0(k_r r)}{dr}\right)dr$$

$$\frac{d}{dr}\left(r\,\frac{dJ_0(k_r r)}{dr}\right)=-k_r^2\,r\,J_0(k_r r)$$

$$=-k_r^2\int_0^{\infty}\bar{p}(r,z)\,J_0(k_r r)\,r\,dr=-k_r^2\,g(k_r,z)$$

$$\int_0^\infty \frac{1}{r}\frac{\partial}{\partial r}\left(r\,\frac{\partial \bar{p}}{\partial r}\right) J_0(k_r r)\, r\, dr = -k_r^2\, g(k_r, z), \tag{10}$$

$$\int_0^\infty \frac{\partial^2 \bar{p}}{\partial z^2}\, J_0(k_r r)\, r\, dr = \frac{d^2}{dz^2}\int_0^\infty \bar{p}(r,z)\, J_0(k_r r)\, r\, dr = \frac{d^2 g(k_r,z)}{dz^2}$$

$$\int_0^\infty k^2(z)\, \bar{p}(r,z)\, J_0(k_r r)\, r\, dr = k^2(z)\, g(k_r, z)$$

$$\int_0^\infty \left[-\frac{\delta(r)\,\delta(z-z_s)}{2\pi r}\right] J_0(k_r r)\, r\, dr = -\frac{\delta(z-z_s)}{2\pi}\int_0^\infty \delta(r)\, J_0(k_r r)\, dr = -\frac{\delta(z-z_s)}{2\pi}\, J_0(0) = -\frac{\delta(z-z_s)}{2\pi}$$

$$-k_r^2\, g(k_r,z) + \frac{d^2 g(k_r,z)}{dz^2} + k^2(z)\, g(k_r,z) = -\frac{\delta(z-z_s)}{2\pi}$$

$$\Longrightarrow \quad \frac{d^2 g(k_r,z)}{dz^2} + \left[k^2(z) - k_r^2\right] g(k_r,z) = -\frac{\delta(z-z_s)}{2\pi}$$

We thus obtain the *depth-separated wave equation*

$$\boxed{\frac{d^2 g(k_r,z)}{dz^2} + \left[k^2(z) - k_r^2\right] g(k_r,z) = -\frac{\delta(z-z_s)}{2\pi}} \tag{11}$$

Now define the vertical wavenumber

$$k_z(z) = \sqrt{k^2(z) - k_r^2}, \qquad \mathrm{Re}\{k_z\} \geq 0,\ \mathrm{Im}\{k_z\} \geq 0, \tag{12}$$

so components with $k_r \leq k(z)$ propagate and those with $k_r > k(z)$ are evanescent. In each homogeneous layer the general solution of (11) is

$$\frac{d^2 \hat{g}}{dz^2} + k_z^2\, \hat{g} = -\frac{\delta(z-z_s)}{2\pi}, \qquad k_z^2 = k^2(z) - k_r^2$$

$$\hat{g}(k_r, z) = C\, e^{ik_z|z-z_s|}$$

$$\frac{d\hat{g}}{dz} = C\,(ik_z)\ \mathrm{sgn}(z-z_s)\, e^{ik_z|z-z_s|}$$

$$\left[\frac{d\hat{g}}{dz}\right]_{z_s^-}^{z_s^+} = C\,(ik_z)\big[1-(-1)\big] = 2ik_z C$$

$$2ik_z C = -\frac{1}{2\pi} \quad \Longrightarrow \quad C = \frac{i}{4\pi k_z}$$

$$\hat{g}(k_r, z) = \frac{i}{4\pi k_z}\, e^{ik_z|z-z_s|}$$

We now bound the domain to a fluid layer of depth D: a pressure-release surface at $z = 0$ (a free surface, where the acoustic pressure vanishes) and a perfectly rigid bottom at $z = D$ (where the vertical particle velocity, and hence $\partial\bar{p}/\partial z$, vanishes). The two conditions are therefore

$$g(k_r, 0) = 0, \qquad \left.\frac{dg}{dz}\right|_{z=D} = 0. \tag{13}$$

The surface condition forces the depth function to a sine that vanishes at $z = 0$, while the rigid bottom forces a cosine that is flat at $z = D$. Imposing both on the general solution fixes the homogeneous coefficients,

$$\frac{i}{4\pi k_z}\, e^{ik_z z_s} + A^+ + A^- = 0,$$

$$\frac{i}{4\pi k_z}(ik_z)\, e^{ik_z(D-z_s)} + A^+(ik_z)\, e^{ik_z D} - A^-(ik_z)\, e^{-ik_z D} = 0,$$

Constructing the point-source Green's function of the Sturm–Liouville problem (11) from the homogeneous solution satisfying each boundary—$\sin(k_z z)$ at the pressure-release surface and $\cos(k_z(D-z))$ at the rigid bottom—joined at the source and normalised by the Wronskian, the depth-dependent Green's function takes the compact standard solution

$$g(k_r, z) = \frac{1}{2\pi k_z}\,\frac{\sin(k_z\, z_<)\cos(k_z(D - z_>))}{\cos(k_z D)}, \qquad z_< = \min(z, z_s), \quad z_> = \max(z, z_s), \tag{14}$$

so that the acoustic field is

$$\bar{p}(r, z) = \int_0^\infty \frac{1}{2\pi k_z}\,\frac{\sin(k_z\, z_<)\cos(k_z(D - z_>))}{\cos(k_z D)}\, J_0(k_r r)\, k_r\, dk_r. \tag{15}$$

The integrand (15) has poles where $\cos(k_z D) = 0$, i.e. at the discrete vertical wavenumbers

$$k_{z,n} = \frac{(n + \frac{1}{2})\pi}{D}, \qquad k_{r,n} = \sqrt{k^2 - k_{z,n}^2}, \qquad n = 0, 1, 2, \ldots \tag{16}$$

Evaluating (15) by residues at these poles collapses the wavenumber integral into a discrete sum of modes, giving the final closed form

$$\boxed{\bar{p}(r, z) = \frac{i}{2D}\sum_{n=0}^{\infty} \sin(k_{z,n} z_s)\sin(k_{z,n} z)\, H_0^{(1)}(k_{r,n} r)} \tag{17}$$

where each term $\psi_n(z) = \sin(k_{z,n} z)$ is a mode of the waveguide, satisfying the surface condition $\psi_n(0) = 0$ and the rigid-bottom condition $\psi_n'(D) = 0$ by construction, and $H_0^{(1)}$ is the outgoing Hankel function carrying the range dependence.

1.1 Data-aided wavenumber-integration formulation

Equation (17) is the residue evaluation of the wavenumber integral (9), valid only when the environment—and hence the kernel $g(k_r, z)$—is fully known. When the environment is unknown but sparse acoustic measurements are available, we do not evaluate the integral. Instead we retain (9) in its integral form and treat the spectral kernel $g(k_r, z)$ itself as the quantity to be learned from data.

The kernel is sampled on a two-dimensional grid: the index $m = 0, 1, \ldots, M-1$ runs over the horizontal-wavenumber samples $k_{r,m} = k_{\min} + m\,\Delta k_r$, and the index $l = 0, 1, \ldots, L-1$ runs over the depth samples z_l at which the field is represented. Each $g(k_{r,m}, z_l)$ is a single complex number, so the set of trainable parameters $\mathcal{T}$—the quantities adjusted during training to fit the measurements—is the full collection of $M \times L$ complex kernel values,

$$\mathcal{T} \equiv \left\{\, g(k_{r,m}, z_l) \,\right\}_{m=0,\, l=0}^{M-1,\, L-1}. \tag{18}$$

The field at any range and depth is synthesised by discretising the wavenumber integral (9) directly. Sampling k_r on the grid $k_{r,m} = k_{\min} + m\,\Delta k_r$ and approximating the integral by a Riemann sum gives the discretised forward model

$$\bar{p}(r, z_l) \approx \sum_{m=0}^{M-1} g(k_{r,m}, z_l)\, J_0(k_{r,m} r)\, k_{r,m}\, \Delta k_r. \tag{19}$$

The finite grid truncates the integral (9): the lower limit $k_{\min}$ discards the interval $[0, k_{\min})$, where the weight $k_r J_0(k_r r)$ is $O(k_r)$ and the integrand contributes negligibly, while the upper limit is chosen so that $k_{r,M-1}$ exceeds $\max_z k(z)$, enclosing every propagating wavenumber (those with $k_r > k$ being evanescent and exponentially small in range). Each term in (19) is the product of a learned complex weight $g(k_{r,m}, z_l)$ and a fixed basis function $J_0(k_{r,m} r)\, k_{r,m}\, \Delta k_r$ determined entirely by the sampling grid; only the weights g are adjusted during training. For a fixed depth z_l, the model is therefore linear in its parameters, and (19) may be written compactly as

$$\bar{\boldsymbol{p}} = \Phi\, \boldsymbol{g}, \qquad \Phi_{jm} = J_0(k_{r,m} r_j)\, k_{r,m}\, \Delta k_r, \tag{20}$$

where $\boldsymbol{g}$ is the vector of trainable kernel values and Φ is a fixed design matrix of Bessel components evaluated at the measurement ranges r_j.

The kernel values (18) are learned by minimising a data-misfit loss. For the ray-basis network in the paper done by Li and Chitre, it minimises

$$L_p(\boldsymbol{r}, y; \mathcal{T}_p) = \big(|\bar{p}(\boldsymbol{r})| - y\big)^2 + \alpha\, \|\boldsymbol{A}\|_1, \tag{21}$$

where $\boldsymbol{A}$ are the ray amplitudes and α controls an L_1 sparsity penalty, summed over the training data. We adopt the same form, with the ray amplitudes replaced by the spectral kernel g,

$$L(\mathcal{T}) = \sum_j \big(|\bar{p}(r_j, z_j)| - y_j\big)^2 + \alpha\, \|g\|_1, \tag{22}$$

over measurements y_j at locations (r_j, z_j).